

Question 1.1

(a) The state is

$$x_t = k_t.$$

The feasible set is

$$\Gamma(x_t) = [0, x_t^\alpha + (1 - \delta)x_t],$$

and the reward function is

$$F(x_t, x_{t+1}) = u(x_t^\alpha + (1 - \delta)x_t - x_{t+1}).$$

Hence the sequence problem is

$$v^*(x_0) = \sup_{\{x_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t F(x_t, x_{t+1}), \quad x_{t+1} \in \Gamma(x_t).$$

k_t is the state because, given k_t , the feasible choices and future utility do not depend on the history before t . If utility also depended on c_{t-1} , then c_{t-1} would also have to be included in the state:

$$x_t = (k_t, c_{t-1}).$$

(b) Starting from the sequence problem,

$$\begin{aligned} v^*(x_0) &= \sup_{x_1 \in \Gamma(x_0)} \left\{ F(x_0, x_1) + \beta \sup_{\{x_{t+2}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t F(x_{t+1}, x_{t+2}) \right\} \\ &= \sup_{x_1 \in \Gamma(x_0)} \{ F(x_0, x_1) + \beta v^*(x_1) \}. \end{aligned}$$

Therefore,

$$v(k) = \sup_{k'} \{ u(k^\alpha + (1 - \delta)k - k') + \beta v(k') \}.$$

Since

$$u_c(c) = c^{-\gamma} \rightarrow \infty \quad \text{as } c \downarrow 0,$$

$c = 0$ cannot be optimal. Moreover, $k' = 0$ would imply zero resources tomorrow, which is incompatible with $c' > 0$. Hence the optimum is interior:

$$0 < k' < k^\alpha + (1 - \delta)k.$$

(c) The Bellman equation is

$$v(k) = \max_{k'} \{ u(k^\alpha + (1 - \delta)k - k') + \beta v(k') \}.$$

The first-order condition for k' is

$$-u_c(c) + \beta v_k(k') = 0,$$

hence

$$u_c(c) = \beta v_k(k').$$

With $k' = g(k)$,

$$v(k) = u(k^\alpha + (1 - \delta)k - g(k)) + \beta v(g(k)).$$

Differentiating with respect to k ,

$$\begin{aligned} v_k(k) &= u_c(c) (\alpha k^{\alpha-1} + 1 - \delta - g_k(k)) + \beta v_k(k') g_k(k) \\ &= u_c(c) (\alpha k^{\alpha-1} + 1 - \delta) + g_k(k) \underbrace{[\beta v_k(k') - u_c(c)]}_{=0 \text{ by FOC}} \\ &= u_c(c) (\alpha k^{\alpha-1} + 1 - \delta). \end{aligned}$$

Therefore,

$$v_k(k) = u_c(c) R(k), \quad R(k) := \alpha k^{\alpha-1} + 1 - \delta.$$

One period ahead,

$$v_k(k') = u_c(c') R(k').$$

Substituting this into the first-order condition gives

$$\boxed{u_c(c) = \beta R(k') u_c(c')}.$$

- (d) The Euler equation says that the marginal utility loss from consuming one unit less today equals the discounted marginal utility gain from saving that unit.

$$R(k') = \alpha (k')^{\alpha-1} + 1 - \delta$$

is the gross marginal return on capital. The term $\alpha (k')^{\alpha-1}$ is the marginal product of capital, while $1 - \delta$ is the fraction of capital that remains after depreciation.

Question 1.2

- (a) For CRRA utility,

$$u_c(c) = c^{-\gamma}.$$

Therefore,

$$u_c^{-1}(x) = x^{-1/\gamma}.$$

Then

$$\begin{aligned} e(k) &= 1 - \frac{u_c^{-1}(\beta R(k') u_c(c'))}{c} \\ &= 1 - \frac{(\beta R(k') (c')^{-\gamma})^{-1/\gamma}}{c} \\ &= 1 - (\beta R(k'))^{-1/\gamma} \frac{c'}{c}. \end{aligned}$$

Hence

$$\boxed{e(k) = 1 - (\beta R(k'))^{-1/\gamma} \frac{c'}{c}}.$$

- (b) At grid point k_i ,

$$k' = k_{g_i}, \quad c_i = h(k_i).$$

Hence

$$\boxed{e_i = 1 - [\beta (\alpha k_{g_i}^{\alpha-1} + 1 - \delta)]^{-1/\gamma} \frac{c_{g_i}}{c_i}}.$$